UNIVERSITY OF SFAX

Faculty of Sciences of Sfax — Computer Science Department

SOFTWARE REQUIREMENTS SPECIFICATION

Final Year Project (PFE) — Academic Year 2025/2026

ATLAS

Aggregated Tunisian Learning & Academic System

An AI-Powered Platform for Centralizing and Democratizing Academic Resources

Prepared by	Mouhamed Gharsallah & Tony Charmant Egerimana
Supervisor	Courinne Amel Zayani
Institution	Faculty of Sciences of Sfax (FSS)
Department	Computer Science & Communication
Academic Year	2025 – 2026
Document Version	V3.1 — March 2026 (English Edition)
Classification	Working Document — Confidential

Table of Contents

Table of Contents 2
1. Context & Problem Statement 4
1.1 Background 4
1.2 Core Problem 4
1.3 Project Objectives 4
2. Solution Overview — ATLAS Platform 5
2.1 Product Vision 5
2.2 Tagline & Identity 5
2.3 Target Users 5
3. Detailed Feature Specifications 6
3.1 User Management & Access Control 6
3.1.1 Role-Based Access Control (RBAC) — 5 Levels 6
3.1.2 Student Self-Registration Flow (Key Feature) 6
3.1.3 Teacher Onboarding (Admin-Initiated Flow) 6
3.1.4 Student Contributor Upgrade Flow 6
3.2 Academic Content Management 8
3.2.1 Multi-Level Course Taxonomy 8
3.2.2 Smart Upload System 8
3.2.3 Document Versioning System 8
3.3 Semantic Search Engine 9
3.3.1 Three-Layer Search Architecture 9
3.3.2 Result Display Logic 9
3.4 AI Module — The Core Differentiator of ATLAS 10
3.4.1 RAG — Retrieval-Augmented Generation (Chat with Your Course) 10
3.4.2 Intelligent Summary Generation 10
3.4.3 Adaptive Flashcard System 10
3.4.4 Quiz Generation & Exam Simulation 10
3.4.5 Auto-Generated Mind Maps 10
3.4.6 Fine-Tuning & Personalization 10
3.5 Community & Forum 11
3.5.1 Structured Academic Forum 11
3.5.2 Reputation & Gamification System 11
3.6 Personalized Dashboards 12
3.6.1 Student Dashboard 12
3.6.2 Teacher Dashboard 12
3.6.3 Admin Dashboard 12
4. Technical Architecture 13

4.1 Architectural Overview 13
4.2 Frontend Stack 13
4.3 Backend Stack 13
4.4 AI & Machine Learning Stack 13
4.5 Database Stack 14
5. Security & Compliance 15
5.1 Authentication & Session Security 15
5.2 Data Security 15
5.3 Content Protection 15
6. Innovative Additional Features 16
6.1 ATLAS Lens — Scan & Digitize 16
6.2 ATLAS Timeline — Exam Schedule Prediction 16
6.3 ATLAS Collaborative — Real-Time Shared Notes 16
6.4 ATLAS Bridge — Semi-Automatic Facebook Import 16
6.5 ATLAS Analytics — Institutional Intelligence Dashboard 16
7. Design & User Experience 17
7.1 Design Principles 17
7.2 Progressive Onboarding 17
7.3 Design System 17
8. Project Planning & Methodology 18
8.1 Methodology 18
8.2 Project Phases 18
8.3 Sprint Review Milestones 18
9. Risk Analysis 19
10. Conclusion & Added Value 20

1. Context & Problem Statement

1.1 Background

The Tunisian higher education system, recognized for producing a significant volume of engineering and scientific talent, is currently experiencing a silent structural crisis: the digital fragmentation of academic resources. With over 450,000 active students, the ecosystem paradoxically relies on informal channels — Facebook groups, expired Google Drive links, and physical copy centers (polycopes) — to distribute knowledge.

This reality was documented and quantified in the research report 'The Digital Archipelago' (Gharsallah & Egerimana, 2026): fragmentation imposes an estimated economic cost of 712 million TND per year, approximately 0.5% of the national GDP.

1.2 Core Problem

The central problem of this project can be summarized as follows:

How can we centralize, organize, validate, and make intelligently accessible all Tunisian academic resources, leveraging AI to personalize learning and eliminate information friction?

1.3 Project Objectives

- Create a centralized hub of academic resources for all Tunisian faculties and departments
- Allow verified teachers to officially publish and manage their course materials
- Enable students to self-register and immediately access the platform's resources
- Provide students with a semantically precise search engine across all disciplines
- Integrate an advanced AI assistant (RAG, Fine-Tuning, NLP) to personalize learning
- Guarantee document permanence beyond graduation — a living institutional memory
- Democratize access to knowledge regardless of the student's socioeconomic background

2. Solution Overview — ATLAS Platform

2.1 Product Vision

ATLAS (Aggregated Tunisian Learning & Academic System) is a full-stack web and mobile platform designed as the missing digital middleware in the Tunisian educational ecosystem. It does not create content — it aggregates, validates, preserves, and makes knowledge intelligent through AI.

ATLAS positions itself at the intersection of three major technological paradigms:

- Knowledge Management Platform (similar to Notion + Academic Wikipedia)
- AI-Powered Learning Assistant (similar to Khanmigo / Perplexity for Education)
- Trust-Based Content Marketplace (similar to StackOverflow + Academia.edu)

2.2 Tagline & Identity

"From chaos to clarity. From archipelago to atlas."

The name ATLAS pays double tribute: to cartography — the tool that transforms scattered islands into a coherent territory — and to mythology, where Atlas carries the entire world on his shoulders, just as the platform carries the knowledge of an entire generation.

2.3 Target Users

Primary Actors	Secondary Actors
Students (Bac, Licence, Master, PhD)	Private Tutors (future scope)
Verified Teachers & University Professors	Parents of Baccalaureate students
Establishment Administrators	Researchers accessing archives
Super-Admin (Platform Level)	Unauthenticated Visitors (preview only)

3. Detailed Feature Specifications

3.1 User Management & Access Control

3.1.1 Role-Based Access Control (RBAC) — 5 Levels

The system implements a Role-Based Access Control model with five distinct privilege levels:

Feature	Description	Tech Stack
Super Admin	Full platform control, manages all establishments and AI configuration	JWT + ACL
Admin (Establishment)	Creates teacher accounts via CSV import, approves student contributors	Multi-tenant
Teacher	Uploads official courses, manages modules and versions, views analytics	Role scope
Student Contributor	Uploads resources pending admin approval, earns special badge and XP	Approval workflow
Student (Standard)	Self-registers, accesses all resources, AI tools, forum, flashcards	Public/Auth

3.1.2 Student Self-Registration Flow (Key Feature)

Students can create their own accounts independently, without any admin intervention. The registration process follows a secure 5-step OTP verification flow:

1. Student fills the registration form: email address, chosen password, faculty, department, and academic level (Licence 1, Master 2, etc.)
2. The system validates the email format and checks for duplicates in real time
3. A 6-digit OTP code is generated (HMAC-SHA1, valid for 24 hours, max 5 attempts) and sent to the student's email via a branded HTML email
4. The student enters the OTP on the /activate page — the account is activated and marked isActive = true
5. The student is immediately redirected to their personalized dashboard, ready to search, download, and use AI tools

This self-service model eliminates the bottleneck of manual account creation and allows any student to onboard in under 2 minutes.

3.1.3 Teacher Onboarding (Admin-Initiated Flow)

Teacher accounts follow a different, admin-controlled flow to ensure institutional verification:

6. The establishment admin uploads a CSV file containing teacher names, email addresses, departments, and assigned modules
7. The system auto-generates a pre-configured Teacher account (isActive = false) for each valid row
8. A personalized invitation email is sent to each teacher with a unique OTP activation link (valid 48 hours)
9. The teacher clicks the link, enters their OTP, sets a password, and completes their professional profile
10. Upon activation, the teacher receives a permanent 'Verified Teacher ✓' badge visible on all their uploads

3.1.4 Student Contributor Upgrade Flow

11. An authenticated student submits a contributor request along with a demonstration document
12. The admin reviews the document quality (OCR score, relevance, clarity)
13. If approved: the student receives the 'Community Contributor' badge and access to the contribution upload dashboard
14. Reputation system: each approved contribution increases the student's credibility score and XP points

3.2 Academic Content Management

3.2.1 Multi-Level Course Taxonomy

Every resource in ATLAS is labeled according to a precise, hierarchical taxonomy:

- Establishment → Department → Major → Level → Module → Chapter
- Resource type: Lecture | Tutorial (TD) | Lab (TP) | Corrected Exam | Summary | Reference Sheet
- Academic year: allows access to archives spanning 5+ previous years
- Language: Arabic | French | English | Bilingual
- AI-generated tags: automatically extracted by NLP (e.g., #quicksort #thermodynamics #OOP)

3.2.2 Smart Upload System

- Drag-and-drop support for PDF, DOCX, PPTX, and high-resolution images
- Automatic OCR (Tesseract + PaddleOCR) to make scanned documents text-searchable
- Duplicate detection by SHA-256 hash + semantic similarity to prevent redundant uploads
- Instant preview before submission — the uploader sees exactly what will be published
- Intelligent compression without loss of readable quality
- Optional watermarking for teachers who wish to protect their intellectual property

3.2.3 Document Versioning System

- Each course has a Git-like version history: V1, V2, V3 with dates and uploaders
- Teachers publish a new version without deleting the previous one — both remain accessible
- Students clearly see 'Version 2025 — Updated' vs 'Archive 2023'
- Visual diff comparison between two versions of the same document

3.3 Semantic Search Engine

3.3.1 Three-Layer Search Architecture

ATLAS implements a hybrid, three-layer search approach for maximum precision:

Feature	Description	Tech Stack
Full-Text Search	MeiliSearch indexes OCR-extracted text from all documents	MeiliSearch
Semantic Search	Multilingual BERT embeddings understand query meaning, not just keywords	FAISS + BERT
Hybrid Fusion	Reciprocal Rank Fusion (RRF) combines both for optimal ranking	RRF Algorithm
Contextual Filters	Filter by department, level, year, type, language, community rating	Faceted Search
Spell Correction	Automatic correction for Arabic, French, and English queries	Custom Dictionary

3.3.2 Result Display Logic

- The most recent official teacher course is always prominently featured (badge 'Official ✓')
- 'Recommended' section: based on student profile, department, and browsing history
- 'Previous Years' section: quick access to archives sorted by academic year
- Visible quality score: community rating + download count + source reliability
- Instant preview: first page of the document visible without requiring a download

3.4 AI Module — The Core Differentiator of ATLAS

This is the primary differentiator of ATLAS. The AI module transforms a file repository into a genuine personalized learning assistant.

3.4.1 RAG — Retrieval-Augmented Generation (Chat with Your Course)

- The student selects one or more official courses, then opens the AI Chat panel
- The system indexes the document into semantic chunks (LangChain + ChromaDB)
- The student asks a question: 'Explain the worst-case complexity of quicksort'
- ATLAS retrieves the most relevant passages and generates a response citing the exact source page
- All answers are anchored in the uploaded document — no hallucinations
- The student sees the exact passage highlighted in the original PDF viewer (full transparency)

3.4.2 Intelligent Summary Generation

- Executive Summary (1 page): the key takeaways of a chapter in 5 bullet points
- Structured Summary: an auto-generated hierarchical outline with headings and sub-headings
- Comparative Summary: compares two versions of a course to identify what changed
- Multilingual Summary: generates a summary in Arabic even if the source course is in French

3.4.3 Adaptive Flashcard System

- Automatic generation of Q&A flashcard pairs from any uploaded course document
- Spaced Repetition Algorithm (SM-2): schedules cards at the optimal moment for memory retention
- Three auto-detected difficulty levels: Definition | Application | Critical Analysis
- Express review mode: 10 cards in 5 minutes for last-minute exam preparation
- Sharing flashcard decks with peers in the same major via a unique public link

3.4.4 Quiz Generation & Exam Simulation

- The AI automatically generates multiple-choice questions (MCQ), True/False, and fill-in-the-blank items
- Exam Simulation mode: configurable timer (30/60/90 min), automatic grading, and detailed correction
- Intelligent feedback: 'You confused X with Y — here is the relevant passage from page 47'
- Performance tracking: score evolution chart over the last 30 days

3.4.5 Auto-Generated Mind Maps

- Interactive graphical visualization of a course's concepts and their interconnections
- Automatic links between chapters sharing common concepts
- Export to PNG / SVG for offline visual revision

3.4.6 Fine-Tuning & Personalization

- Base model: Mistral 7B or LLaMA 3 fine-tuned on a Tunisian academic corpus
- Adapted to the vocabulary of Tunisian exams (terminology from FSS, FST, ENIT, etc.)
- Learning from past exam patterns to better target revision recommendations
- Student learning profile: detects knowledge gaps and suggests targeted resources

3.5 Community & Forum

3.5.1 Structured Academic Forum

The ATLAS forum is organized by module and establishment — not chronologically:

- Each course has its own linked discussion space
- Questions are tagged: Comprehension | Exercise | Exam | Missing Resource
- StackOverflow-style voting system: the best answer automatically rises to the top
- Teachers can pin official answers to provide authoritative guidance
- Anti-spam system: rate limiting and AI moderation to detect cheating requests

3.5.2 Reputation & Gamification System

Feature	Description	Tech Stack
XP Points	Earned by uploading, answering, and voting. Unlock platform features.	Leaderboard
Badges	Contributor, Expert, Mentor, Pioneer, Quiz Champion...	Achievement system
Levels	Student → Contributor → Expert → Mentor (4 levels)	Progression
Anonymous Ranking	Top contributors per major (opt-in anonymization, GDPR-friendly)	Privacy-first

3.6 Personalized Dashboards

3.6.1 Student Dashboard

- Activity feed: new courses in their major, forum replies, flashcard review reminders
- Study progress: % of chapters studied, average quiz score, number of cards mastered
- Exam calendar: importable to Google Calendar / iCal with push notification reminders
- 'Start Reviewing' CTA: direct link to flashcard decks due for review today
- Complete history: all documents viewed, downloaded, and annotated with timestamps

3.6.2 Teacher Dashboard

- Course analytics: number of views, downloads, and forum questions per course
- Temporal heatmap: peak consultation periods (students access more before exams)
- Automatic AI insight: 'This chapter generates the most questions — students struggle with recursion'
- Management of contribution requests linked to their modules
- Notifications: when a student shares a correction of one of their past exams

3.6.3 Admin Dashboard

- Global overview: number of users, pending uploads, reports, usage statistics
- CSV teacher import + bulk OTP invitation sending with delivery tracking
- Centralized moderation: approve/reject contributions with comment and XP attribution
- Multi-tenant establishment management with isolated data scopes

4. Technical Architecture

4.1 Architectural Overview

ATLAS adopts a modular microservices architecture, deployable on private or public cloud, designed to handle 450,000+ potential users. All components communicate via REST APIs with asynchronous task queues for heavy processing (OCR, AI inference, email dispatch).

4.2 Frontend Stack

Feature	Description	Tech Stack
Framework	Next.js 14 (App Router, SSR/SSG for SEO performance)	React 18 + TS
UI / Design	Tailwind CSS + shadcn/ui — coherent ATLAS design system	TypeScript strict
State	Zustand (global auth state) + TanStack Query (server state)	TanStack v5
PDF Viewer	PDF.js with RAG source highlight and page scroll sync	Mozilla PDF.js
Mind Maps	React Flow / D3.js for interactive concept graph visualizations	SVG / Canvas
Flashcards	Framer Motion for 3D flip card animations at 60fps	CSS 3D + Motion
PWA	Offline access to downloaded courses, push notifications for reviews	Service Worker

4.3 Backend Stack

Feature	Description	Tech Stack
API Gateway	FastAPI (Python) — high performance, auto Swagger documentation	FastAPI + Uvicorn
Auth Service	JWT Access Token (15 min) + Refresh Token (7 days) + OTP (PyOTP)	OAuth2 / bcrypt
File Service	Upload, compression, OCR trigger, versioning of documents	MinIO (S3-compatible)
Search Service	MeiliSearch (full-text) + pgvector HNSW index (semantic)	Hybrid RRF Fusion
Notification	Transactional email (Resend), Push (Firebase FCM), In-app WebSocket	Async / Celery
Task Queue	Celery + Redis for async processing (OCR, AI generation, emails)	Celery / Redis 7
Real-time	WebSocket for live forum updates, notifications, collaborative presence	Socket.io / FastAPI WS

4.4 AI & Machine Learning Stack

Feature	Description	Tech Stack
LLM Base	Mistral 7B or LLaMA 3.1 — self-hosted, no external API dependency	Ollama / vLLM
Fine-Tuning	QLoRA on Tunisian academic corpus (courses + scanned exams)	PEFT / HuggingFace
RAG Pipeline	LangChain + ChromaDB — retrieval-augmented on uploaded documents	LangChain v0.3
Embeddings	sentence-transformers multilingual BERT (FR/AR/EN)	SBERT
OCR	PaddleOCR (Arabic + French) + Tesseract for degraded documents	PaddleOCR v2
Classification	Auto-classify document type and academic level (Random Forest)	scikit-learn
Recommendation	Collaborative filtering (SVD) + content-based for course suggestions	Surprise lib
NLP Tagging	KeyBERT for automatic academic keyword extraction from OCR text	KeyBERT
Duplicate Detection	SimHash + cosine similarity to prevent redundant uploads	MinHash / SimHash
Quiz Generation	T5/BART fine-tuned on Tunisian exam question patterns	Seq2Seq / Beam search

4.5 Database Stack

Feature	Description	Tech Stack
PostgreSQL 16	Primary relational database — users, courses, metadata, transactions	pgvector extension
pgvector	Vector storage and HNSW ANN queries for semantic search	HNSW index
Redis 7	Session cache, rate limiting, search result caching, Celery broker	Redis v7
MinIO	S3-compatible object storage for all uploaded files (PDF, images)	Self-hosted S3
ChromaDB	Dedicated vector database for the RAG pipeline document chunks	Python-native

5. Security & Compliance

5.1 Authentication & Session Security

- JWT Access Token (15 min expiry) + Refresh Token (7 days, one-time-use with automatic rotation)
- OTP via email for student account activation (6-digit HMAC-SHA1 code, 24-hour TTL, max 5 attempts)
- OTP via email for teacher onboarding (48-hour TTL, single-use, linked to establishment domain)
- Rate limiting per IP and per account to prevent brute-force attacks (Redis counter)
- Optional 2FA for Teacher and Admin accounts (TOTP via Google Authenticator)
- Session management: users can remotely log out of all devices simultaneously

5.2 Data Security

- Password hashing: bcrypt with cost factor 12 — industry standard, resistant to GPU attacks
- Encryption of sensitive files at rest (AES-256) in MinIO
- HTTPS mandatory across all endpoints — TLS 1.3 minimum
- CORS strictly configured by allowed origin domain
- Full input validation and sanitization (SQL injection, XSS, CSRF protection)

5.3 Content Protection

- Community reporting system: users flag inappropriate, misleading, or infringing content
- Invisible steganographic watermark on all downloaded documents for traceability
- Anti-scraping protection: rate limiting + CAPTCHA on public endpoints
- Clear copyright policy: teachers retain all intellectual property rights over their uploads

6. Innovative Additional Features

6.1 ATLAS Lens — Scan & Digitize

A mobile companion feature (PWA) that allows scanning of physical polycope documents:

- The student points their camera at a paper document
- Real-time OCR with automatic correction for distortion and poor lighting
- Extracted text is formatted and proposed for upload to ATLAS instantly
- Automatic department and module detection from the content analysis
- Direct contribution from scan — ending the cycle of 'photocopy → blurry scan → MediaFire'

6.2 ATLAS Timeline — Exam Schedule Prediction

An intelligent calendar visualization built from previous years' uploaded exam documents:

- The AI analyzes historical exam dates and patterns from archived documents
- Generates a predicted exam schedule for the current session
- Alerts 2 weeks before the predicted exam period: 'Start reviewing Algorithms now'
- One-click import to Google Calendar / iCal

6.3 ATLAS Collaborative — Real-Time Shared Notes

A collaborative editor (Notion-style) linked to each course document:

- Students from the same major annotate a shared document together in real time
- Annotations are linked to specific page numbers and text passages of the course
- Full version history of annotations — traceability of who added what and when
- Teachers can view the annotation layer to understand student difficulties

6.4 ATLAS Bridge — Semi-Automatic Facebook Import

A migration tool to combat the Facebook Trap:

- Interface for manually importing files from existing Facebook study groups
- User pastes the post URL, ATLAS fetches the attached file automatically
- AI auto-categorization + submission to admin validation workflow
- Progressive reduction of dependency on informal, ephemeral Facebook groups

6.5 ATLAS Analytics — Institutional Intelligence Dashboard

For faculties, research coordinators, and the Ministry (future scope):

- Which courses generate the most questions? → detects systemic pedagogical gaps
- Which resources are most consulted before exams? → validates content relevance
- Mapping access inequalities by region and establishment
- Automatic monthly PDF report export for faculty deans

7. Design & User Experience

7.1 Design Principles

- Mobile-First: 70%+ of Tunisian students access the web via smartphone on 4G connections
- Accessibility WCAG 2.1 AA: sufficient contrast, keyboard navigation, screen reader support
- Native Dark Mode: for late-night study sessions without eye strain
- Multilingual UI: French / Arabic (full RTL layout) / English — seamlessly switchable
- Performance targets: LCP < 2.5s, FID < 100ms, CLS < 0.1 (Core Web Vitals)

7.2 Progressive Onboarding

- Student registration in under 30 seconds: email + password + faculty + level
- Interactive guided tour on first login (maximum 5 steps, skippable)
- Engaging empty states: popular course suggestions before the first search
- Gamified profile completion progress bar to encourage full setup

7.3 Design System

- Palette: Academic Blue (#1B3A6B) + White + Accent Cyan (#2E86C1)
- Typography: Inter for Latin script, Noto Naskh Arabic for Arabic script
- Components: customized shadcn/ui + proprietary ATLAS component library
- Icons: Lucide React — total visual consistency across the platform

8. Project Planning & Methodology

8.1 Methodology

The project adopts an Agile methodology adapted to the academic context:

- 2-week sprints with regular demonstrations to the academic supervisor
- Prioritized backlog by functional value and technical complexity (MoSCoW method)
- Continuous testing: unit tests (pytest), integration tests (Playwright E2E), load tests (Locust)
- CI/CD with GitHub Actions — automatic deployment to staging on each merged PR

8.2 Project Phases

Feature	Description	Tech Stack
Phase 0 — Setup (2 wks)	Dev environment, database schema, UI mockups, final tech choices	Figma / Docker
Phase 1 — Core (6 wks)	Auth (student self-register + teacher OTP), upload, full-text search, RBAC	MVP deliverable
Phase 2 — AI (6 wks)	RAG pipeline, summaries, flashcards SM-2, OCR, embeddings, quiz generation	LangChain / BERT
Phase 3 — Community (4 wks)	Forum, reputation system, contributions, real-time notifications	WebSocket / XP
Phase 4 — Polish (4 wks)	Analytics, mobile PWA, load testing, optimizations, LLM fine-tuning	QLoRA / Locust
Phase 5 — Defense (2 wks)	Final documentation, PFE report, demo preparation, presentation	LaTeX / Beamer

8.3 Sprint Review Milestones

Technical Milestones	Delivery Milestones
Sprint 1 (Mar 21): Auth + Upload functional	Sprint 2 (Apr 11): Semantic search + OCR pipeline
Sprint 3 (Apr 26): RAG Chat operational on PDF	Sprint 4 (May 2): Forum + Gamification + Dashboards
Sprint 5 (May 23): Final PFE Defense	

9. Risk Analysis

Feature	Description	Tech Stack
Insufficient GPU resources	Fine-tuning requires GPUs. Mitigation: Google Colab Pro, FSS partnership, or Q4 quantization.	High Risk
Arabic OCR quality	Arabic OCR is imperfect. Mitigation: correction pipeline + community contribution for fixes.	Medium Risk
Teacher adoption	Resistance to change. Mitigation: ultra-simple UX + ROI demonstration via usage analytics.	Medium Risk
Copyright compliance	Protected content uploads. Mitigation: clear policy, reporting system, uploader responsibility.	Legal Risk
Scalability at 450K users	High concurrent load. Mitigation: microservices architecture + load balancing + CDN.	Technical Risk
LLM hallucination	AI generates false information. Mitigation: strict RAG grounding with similarity threshold cutoff.	AI Risk

10. Conclusion & Added Value

ATLAS represents far more than a final year project. It is a concrete technological response to a documented crisis affecting hundreds of thousands of young Tunisians. By combining the latest advances in AI (RAG, Fine-Tuning, multilingual NLP) with a user-centered approach — including frictionless student self-registration — and a rigorous governance model, ATLAS has the potential to structurally transform access to knowledge in Tunisia.

This project demonstrates mastery of the most sought-after technologies in the current job market: Large Language Models, microservices, vector search engines, modern full-stack development, and complex multi-stakeholder system management. It constitutes a first-rate technical showcase for the professional insertion of its authors.

"From chaos to clarity. From archipelago to atlas."

Document prepared by Mouhamed Gharsallah & Tony Charmant Egerimana
Faculty of Sciences of Sfax — PFE 2025/2026 — Version 3.1 — March 2026